

Replication Package for: The moment is here: a generalised class of estimators for fuzzy regression discontinuity designs

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1 Overview

This package contains all code necessary to replicate the Monte Carlo simulations and empirical application presented in “The moment is here: a generalised class of estimators for fuzzy regression discontinuity designs”. To produce all outputs in the paper, simply run `master.R` (recommended from RStudio, otherwise explicitly set your working directory by uncommenting `setwd()` on line 16 of `master.R`).

2 License

This replication package is made available under the MIT License. See `LICENSE.txt` file for details.

3 Files

3.1 Main files:

The replication package contains the following directories and files:

Root Directory:

- `README.pdf` - This file
- `config.R` - Main configuration file
- `master.R` - Master script to run everything
- `LambdaFRD.R` - Computes lambda-class estimators
- `LICENSE.txt` - License file

`simulations/` - Monte Carlo simulation code

- `simulations_main.R` - Main simulation loop
- `simulation_utils.R` - Utility functions for simulations
- `estimator_summary.R` - Estimator performance summaries
- `latex/` - LaTeX conversion scripts
 - `convert_simulation_csv_to_latex.py` - Main simulation csv-to-latex file
 - `simulation_latex_utils.py` - Simulation csv-to-latex utility functions

`application/` - Angrist-Lavy (1999) class-size effects replication

- `application_main.R` - Main application script
- `application_utils.R` - Utility functions for applications
- `data/` - Data folder
 - `anglavy99_data.csv` - Application data
- `latex/` - LaTeX conversion scripts
 - `convert_application_csv_to_latex.py` - Main application csv-to-latex file
 - `application_latex_utils.py` - Application csv-to-latex utility functions

3.2 Generated files:

After running `master.R`, the following directories and intermediate files will be generated:

- `simulations/output/` - Simulation results (intermediate .csv files before LaTeX tables)
 - `error_tracking_summary` - Tracks simulation repetitions that fail
 - `estimator_summary.csv` - Raw data for Table 5.2
 - `simulation_results_dgp1_beta.csv` - Raw data for Tables D.2 and D.4
 - `simulation_results_dgp1_normal.csv` - Raw data for Tables 5.1 and 5.3
 - `simulation_results_dgp1_uniform.csv` - Raw data for Tables D.1 and D.3
 - `simulation_results_dgp2_beta.csv` - Raw data for Tables D.7 and D.10
 - `simulation_results_dgp2_normal.csv` - Raw data for Tables D.5 and D.8
 - `simulation_results_dgp2_uniform.csv` - Raw data for Tables D.6 and D.9
- `application/anglavy99_output.csv` - Raw data for Tables 6.1 and D.11

The intermediate files will be converted into LaTeX tables in the generated folder `output/`:

- **output/** - Final outputs for paper
 - **tex_tables/** - All tables (LaTeX format)
 - * `summary_table.tex` - .tex code for Table 5.2
 - * `table_anglavy.tex` - .tex code for Table 6.1
 - * `table_estimation_dgp1_beta.tex` - .tex code for Table D.2
 - * `table_estimation_dgp1_normal.tex` - .tex code for Table 5.1
 - * `table_estimation_dgp1_uniform.tex` - .tex code for Table D.1
 - * `table_estimation_dgp2_beta.tex` - .tex code for Table D.7
 - * `table_estimation_dgp2_normal.tex` - .tex code for Table D.5
 - * `table_estimation_dgp2_uniform.tex` - .tex code for Table D.6
 - * `table_inference_dgp1_beta.tex` - .tex code for Table D.4
 - * `table_inference_dgp1_normal.tex` - .tex code for Table 5.3
 - * `table_inference_dgp1_uniform.tex` - .tex code for Table D.3
 - * `table_inference_dgp2_beta.tex` - .tex code for Table D.10
 - * `table_inference_dgp2_normal.tex` - .tex code for Table D.8
 - * `table_inference_dgp2_uniform.tex` - .tex code for Table D.9

4 Data

All simulation data are generated by the code provided in this replication package. To ensure reproducibility, seeds are set in the following way within the parallel loop:

```
param_id <- dgp * 10000 + running_variable * 1000 + setup * 100 +
  round(p1 * 10) + (n %/% 100)
iter_seed <- 1234 + param_id * NSIM + i
set.seed(iter_seed)
iter_seeds[i] <- iter_seed
```

This architecture ensures that:

- Results from any parameter configuration can be exactly replicated without running the entire grid
- Each simulation draw has a unique, deterministic seed
- Parallel processing does not affect reproducibility

4.0.1 Important Notes

- **Exact replication requires:** NSIM = 10000 (as used in the paper)
- **Changing NSIM:** Modifying NSIM in `config.R` will change the seed sequence and produce different (but statistically equivalent) results.

4.0.2 Secondary Data (Empirical Application):

- Angrist and Lavy (1999) class-size effects on test scores data
- Reference: Angrist, J. D., & Lavy, V. (1999). Using Maimonides' rule to estimate the effect of class size on scholastic achievement. *The Quarterly journal of economics*, 114(2), 533-575.
- Available at: <https://economics.mit.edu/people/faculty/josh-angrist/angrist-data-archive>
- Data come cleaned, no restrictions on access

5 Computational Requirements

5.1 Software Requirements

R (version 4.3.1 or higher)

Required R packages (automatically installed by `master.R` if missing):

- `dplyr` - Data manipulation
- `haven` - Reading Stata .dta files
- `rdrobust` - RD robust estimation/inference procedures
- `RDHonest` - Honest confidence intervals for RD designs
- `parallel` - Parallel computing support
- `doParallel` - Parallel backend for foreach
- `foreach` - Parallel loop construct
- `expm` - Matrix exponential and related functions

Python (version 3.8 or higher)

Required Python packages (automatically installed by `master.R` if missing):

- `pandas` ($\geq 2.0.0$) - Data manipulation and CSV processing

5.2 Hardware Requirements

- **Memory:** Minimum 16GB RAM recommended for full simulations
- **Storage:** Approximately 2GB disk space required
- **Processor:** Multi-core processor recommended (simulations use parallel processing)
- **Operating System:** Windows 10/11, macOS 10.14+, or Linux

5.3 Runtime

The full replication package took approximately 10 hours to run locally on a 24-core machine with 32GB RAM, with 23 cores used in parallel for Monte Carlo simulations (implemented via `fork` / `mclapply`), which dominated runtime (~9 hours).

Note: Runtime can be significantly reduced by:

- Modifying `NSIM` and `NSIM_FIGURES` in `config.R` (at the cost of Monte Carlo accuracy)
- Modifying `GRID_POINTS` for bias-aware AR confidence intervals in the application
- Using more CPU cores (modify `NUM_CORES` in `config.R`)
- Reducing the size of the (`DGPS`, `RUNNING_VARIABLES`, `SETUPS`, `P1_VALUES`, `NS`)' grid

5.4 Dependencies Installation

All required R and Python packages are automatically installed when running `master.R`. Manual installation is not necessary but can be done as follows:

R packages:

```
install.packages(  
  c("dplyr", "haven", "rdrobust", "RDHonest", "parallel", "doParallel",  
    "foreach", "triangle", "expm")  
)
```

Python packages:

```
pip install "pandas>=2.0.0"
```